

School of Computer Science Engineering and Technology

Lab: 3

Course	B. Tech.
Type	Elective
Course Code	CSET346
Course Name	NATURAL LANGUAGE PROCESSING
Year	2026
Semester	Odd
Batch	2026-27

CO-Mapping

Question	CO1	CO2	CO3
Q1	√		
Q2		√	
Q3		√	√
Q4			√

Statistical Language Modeling and Sequence Analysis

Objective:

To develop practical understanding of Language Corpora, N-gram Language Models, Hidden Markov Models, Ambiguity, Coreference, and Domain-Specific Language using Python and standard NLP libraries.

Problem Statement

Use suitable language corpora and Python libraries to analyze sequential patterns, construct statistical language models, implement HMM-based sequence tagging, and investigate linguistic challenges.

Datasets / Resources

- NLTK Brown Corpus – corpus analysis and HMM POS tagging.
- NLTK Gutenberg Corpus – N-gram language modeling.
- 20 Newsgroups Dataset – domain-specific language analysis.
- Manually created contextual sentences – ambiguity and coreference analysis.

Instructions

- Use Python with NLTK, pandas, NumPy, and scikit-learn where appropriate.
- Download required corpora before execution.
- Display intermediate and final outputs.
- Include comments explaining major implementation steps.

Q1. Use the NLTK Brown Corpus to perform corpus analysis and generate N-gram statistics.

- (a) Load the Brown Corpus and select the news category.
- (b) Extract at least 5,000 tokens and perform basic preprocessing.
- (c) Calculate total tokens and vocabulary size.
- (d) Generate unigram, bigram, and trigram sequences.

- (e) Display the ten most frequent examples of each.
- (f) Explain how a language corpus supports statistical language analysis.

Expected Output: Corpus statistics, vocabulary size, N-gram frequencies, and interpretation.

Q2. Construct a Bigram Language Model using the NLTK Gutenberg Corpus.

- (a) Load a suitable English text from Gutenberg.
- (b) Tokenize and normalize the corpus.
- (c) Construct unigram and bigram frequency counts.
- (d) Implement add-one smoothing.
- (e) Calculate scores for at least three test sentences.
- (f) Explain the importance of smoothing.

Expected Output: Vocabulary size, N-gram counts, sentence scores, and observations.

Q3. Implement a Hidden Markov Model for Part-of-Speech sequence tagging.

- (a) Load the Brown Corpus with universal POS tags.
- (b) Divide tagged sentences into training and testing sets.
- (c) Train an HMM-based tagger.
- (d) Predict tags for at least five unseen sentences.
- (e) Calculate test accuracy.
- (f) Explain hidden states, observations, transition probabilities, and emission probabilities.

Expected Output: Training/testing information, predicted POS tags, accuracy, and explanation of HMM components.

Q4. Investigate ambiguity, coreference, and domain-specific language using contextual examples and the 20 Newsgroups dataset.

- (a) Create at least five ambiguity examples using words such as bank, bat, light, or match.
- (b) Identify intended meanings using contextual clues.
- (c) Create at least five coreference examples and identify antecedents.
- (d) Load sci.space and comp.graphics from 20 Newsgroups.
- (e) Extract and display at least 20 frequent domain-specific terms.
- (f) Explain how ambiguity, coreference, and domain vocabulary affect NLP performance.

Expected Output: Ambiguity examples, contextual clues, coreference relations, domain vocabulary, and observations.

